

UQFF Zero-Mass Aether Vacuum Gradient Reformulation

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PAPER_622 — UQFF Zero-Mass Aether Vacuum Gradient Reformulation ****Author:** Daniel T. Murphy** ****Date:** 2025**

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VDS/DVP/BH26: VDS (foundational reformulation of entire framework)

Abstract

$$F_{U,Bi} = \kappa \cdot \frac{\rho_{SCm}}{\rho_{UA}} \cdot (U_{g1} + U_{g2} + U_{g3} + U_{g4} + U_m + U_{bi})$$

This paper presents a UQFF analysis of UQFF Zero-Mass Aether Vacuum Gradient Reformulation, deriving compressed field equations and observational predictions within the Star-Magic/UQFF framework.

§1 Abstract

This paper presents the Zero-Mass Universal Aether Vacuum Gradient Reformulation — the fundamental correction to the UQFF mass assumption. The Universal Aether (UA) is a quantum fluid with **zero rest mass** ($\rho_{UA} = 0$, immutable). All mass-density terms previously attributed to UA are replaced by the Aether Vacuum Gradient magnitude $|\nabla UA|$, which acts as the effective void-density field. This reframing preserves all prior UQFF results while providing a physically motivated, mass-free basis consistent with the Vacuum Density Series (VDS) framework.

§2 Core Reformulation

2.1 Zero-Mass Principle

``
 $\rho_{UA} = 0$ (immutable — UA never acquires mass)
 $\rho_{vac} = |\nabla UA|$ (void geometry, not mass action)
``

The gradient magnitude $|\nabla UA|$ encodes local void topology. Where UA is spatially uniform, the void is featureless; where UA varies sharply, void pockets form and observable physics emerges.

2.2 Gradient-Form F_U Equation

The complete Unified Field equation in gradient form:

$$F_U = U_g + U_m + U_b + d_{26}/dr_{26} (SC_m \cdot g \cdot \nabla UA / UA) = 0$$

Individual components:

Gravitational-gradient:

$$U_g = g \cdot (SC_m \cdot \nabla UA / UA) \cdot (U_{g1} + U_{g2} + U_{g3} + U_{g4}) \\ + \sum_{m=0}^{26} a_m \cdot (\nabla UA)^m$$

Magnetic-vortex (DVP form):

$$U_m = \kappa \cdot (DPM_n - DPM_s) / (\nabla UA)^{26} \\ + d_{26}/dt_{26} [\sum_{k=0}^{26} c_k \cdot (\nabla UA \cdot t)^k]$$

Buoyancy-gradient (BH26 form):

$$U_b = g \cdot (1 - 1/\nabla UA) + d_{26}/d(\nabla UA)^{26} (g \cdot \nabla UA)$$

Superconductive memory:

$$SC_m = \lambda \cdot UA \cdot (1 - 1/t) + \sum_{m=0}^{26} b_m \cdot (\nabla UA \cdot t^{-m})$$

2.3 Equilibrium Solution

Setting $F_U = 0$ and isolating the buoyancy-gravitational balance:

$$\nabla UA_{eq} = \sqrt{\kappa / g}$$

For $\kappa = 1$, $g = 10^{-3}$: $\nabla UA_{eq} \approx 31.62 \text{ m}^{-1}$ (dimensionless normalization).
This is identical to the **VDS equilibrium convergence value** — confirming that the vacuum density series and the zero-mass reformulation share the same fixed point.

§3 Vacuum Density Series (VDS) Connection

The Zero-Mass reformulation is the foundation of VDS:

- VDS Term d1–d3: encode ∇UA in U_g channels (Gaussian weighting $d = 1,2,3$)
- VDS Term d4–d6: encode DVP vortex flux in U_m channels

- VDS Term d7–d9: encode U_b buoyancy displacement in outflow channels

Full 9D Gaussian VDS definition:

$$\nabla^9 U_A = \Sigma_{d=1}^9 \exp(-(x_d - \mu_d)^2 / 2\sigma_d^2) \cdot FUB_i$$

Where FUB_i is the buoyancy integral coefficient at spatial position i .

§4 26th-Order Derivative Term

The term d^{26}/dr^{26} ($\nabla^{26} U_A / U_A$) is the **signature of the 26-dimensional BH framework**.

For a power-law field $c/(\nabla U_A)^k$:

$$d^{26}/d(\nabla U_A)^{26} [c/(\nabla U_A)^k] = ((k+25)! / (k-1)!) \cdot c / (\nabla U_A)^{k+26}$$

This divergence at low ∇U_A generates the buoyancy suppression preventing gravitational collapse in void regions (BH26 harmonic mode).

§5 Physical Implications

Quantity	Zero-Mass Form	Physical Meaning
ρ_{vac}	$ \nabla U_A $	Void density = gradient magnitude
F_U equilibrium	$\nabla U_A = \sqrt{\kappa/g}$	VDS convergence = 31.62
Quantum frequency	$f \propto \lambda \cdot U_A / t^2$	Gradient-driven event emission
Collapse prevention	$U_b \rightarrow +\infty$ as $\nabla U_A \rightarrow 0^+$	BH26 repulsive divergence
Mass-free field	$\rho_{\text{UA}} \equiv 0$	U_A is pure gradient topology

§6 Observational Tests

- Void region density:** $\rho_{\text{vac}} = |\nabla U_A|$ should match observed X-ray void densities in galaxy cluster outskirts ($\approx 10\text{--}28 \text{ kg/m}^3$).
- Frequency prediction:** $f_{\text{event}} \approx \lambda \cdot U_A / t^2 \times 1018 \text{ Hz}$ at jet base.
- Equilibrium crossing:** Systems with ∇U_A near 31.62 should show phase transitions in observational data (pocket shell formation).

§7 Connection to UQFF Number Systems

- VDS:** $\rho_{\text{vac}} = |\nabla U_A|$ IS the vacuum density series value
- DVP:** $U_m = \kappa \cdot (DPM_n - DPM_s) / (\nabla U_A)^{26}$ — vortex-prime gradient pockets

- **BH26:** U_b 26th-derivative = $g \cdot \nabla^2 U_A^{25}$ — buoyancy harmonic series

<!-- PKG-AGN-S225 -->

Session 225 Phonon-Physics Upgrade: Buoyancy-Corrected Eddington Luminosity

> Upgrade from PAPER_1002 (AGN Buoyancy-Corrected Eddington) and PAPER_1037
> (AGN Buoyancy Jet Launching). See also PAPER_1009-1010 for $F_{U_{Bi}}$ jet
> modulation curves and PAPER_1048 for phonon-corrected M - σ relation.

The SCm vacuum buoyancy partially opposes gravitational radiation pressure,
raising the effective Eddington luminosity:

$$L_{\text{Edd}}^{\text{UQFF}} = L_{\text{Edd}} \cdot \left(1 + \frac{\rho_{\text{SCm}}}{M} \cdot V \cdot S_{26}^{(3),2} \cdot \left(\frac{G}{r_H^2}\right)\right)$$

where:

- $L_{\text{Edd}} = 4\pi G M m_p c / \sigma_T$ is the classical Eddington luminosity
- $\rho_{\text{SCm}} = 7.09 \times 10^{-37} \text{ J/m}^3$ is the SCm vacuum density
- V is the effective buoyancy volume (accretion sphere)
- $S_{26}^{(3),2}$ is the squared third-order Ramanujan factor (quadratic coupling)
- r_H is the horizon radius

Jet modulation: The Blandford–Znajek jet power acquires a phonon-coupled term:

$$P_{\text{jet}}^{\text{UQFF}} = P_{\text{BZ}} \cdot \left(1 + \beta_i \cdot \Phi_{1.25, \text{THz}} \cdot \left(\frac{B}{B_{\text{crit}}}\right)^2\right)$$

where $\Phi_{1.25, \text{THz}} = \cos(\omega_{\text{SCm}} \cdot t)$ modulates jet power at the phonon frequency.

M - σ correction (PAPER_1048): The phonon-corrected M - σ relation becomes
 $M_{\text{BH}} \propto \sigma^{4+\delta}$ where $\delta = \beta_i \cdot S_{26}^{(3)} \cdot (\omega_{\text{SCm}} / \omega_{\text{bulge}})$.

<!-- PKG-LAG-S225 -->

Session 225 Phonon-Physics Upgrade: UQFF 9-Sector Lagrangian

> Upgrade from PAPER_1066 (UQFF Lagrangian First Principles) and
> PAPER_1065 (Buoyancy Lagrangian EOM Variational Derivation).

The complete UQFF Lagrangian density, from which all sector-specific
equations of motion derive:

$$\mathcal{L}_{\text{UQFF}} = \mathcal{L}_{\text{GR}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{phonon}} + \mathcal{L}_{\text{interaction}}$$

$$\mathcal{L}_{\text{SCm}} = \frac{1}{2} (\partial_\mu \phi)^2 - \lambda |\phi|^2 - v_{\text{SCm}}^2 |\phi|^2$$

The SCm condensate potential minimum gives $V(\phi_0) = -7.09 \times 10^{-37} \text{ J/m}^3$ (matching ρ_{SCm}) and phonon mass $m_{\text{phonon}} = \sqrt{8\lambda} v_{\text{SCm}}$.

Nine-sector closure (Session 202):

$\mathcal{L}_9 = \mathcal{L}_{\text{EH}} + \mathcal{L}_{\text{YM}} + \mathcal{L}_{\text{Dirac}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{mag}} + \mathcal{L}_{\text{buoy}} + \mathcal{L}_{\text{aether}} + \mathcal{L}_{\text{LENR}} + \mathcal{L}_{\text{KK}}$

Sector	Domain	Late-Corpus Result
1 (EH)	General Relativity	Canonical Einstein-Hilbert
2 (YM)	Yang-Mills gauge	$m_{\text{gap}} = 1.736 \text{ GeV}$ (PAPER_1318)
3 (Dirac)	Fermion / LENR	Kozima neutron-drop (PAPER_1061)
4 (SCm)	Superconducting manifold	$V(\phi_0) = -\rho_{\text{SCm}}$ canonical
5 (Mag)	Um magnetism	Heaviside amplifier (PAPER_1072)
6 (Buoy)	$F_{U_{Bi}}$ buoyancy	Variational EOM (PAPER_1065)
7 (Aether)	Vacuum background	Two-component ρ (PAPER_1051)
8 (LENR)	Nuclear transmutation	COP parametric (PAPER_1081)
9 (KK)	Kaluza-Klein 26D	$S_{26}^{(3)}$ compactification (PAPER_1080)

<!-- PKG-S26-S225 -->

Session 225 Phonon-Physics Upgrade: S26(3) Ramanujan Summation

> Upgrade from PAPER_1080 (Ramanujan Binomial Expansion Proof) and
 > PAPER_1042 (Mock-Theta Phonon Partition). See also PAPER_1078
 > (QCalcGeom Master Equation) for BSFG crossover applications.

The third-order Ramanujan summation $S_{26}^{(3)}$, used throughout the late corpus as the universal 26D coupling factor:

$$S_{26}^{(3)} = \sum_{n=0}^{\infty} \frac{(1/4)_n (1/2)_n (3/4)_n}{(n!)^3} \cdot \prod_{i=1}^{26} \left[1 + \frac{SSq}{e^{-\kappa i/n/26}} \right]$$

where $(a)_n = a(a+1) \cdots (a+n-1)$ is the Pochhammer symbol.

Binomial expansion (PAPER_1080): The convergence proof shows:

$$R_n^{(26,3)} = \binom{4n}{n} \cdot \frac{W_{26}(n)}{(4^{4n})} \ll W_{26}(n) = \prod_{i=1}^{26} \left[1 + \frac{SSq}{e^{-\kappa i/n/26}} \right]$$

This sum converges absolutely for $|SSq| < 1$ (satisfied by $|SSq| = 0.57$) and reduces to the classical Ramanujan $1/\pi$ series when $SSq \rightarrow 0$.

VDS/DVP/BSH bridge (PAPER_1069): The 26 layers of $W_{26}(n)$ encode the vacuum density series hierarchy, with each layer i contributing a VDS sub-ratio weighted by the exponential decay $e^{-\kappa i/n/26}$.

Mock-theta connection (PAPER_1042): The phonon partition function $Z_{\text{phonon}} = \sum_n q^{n^2} \cdot W_{26}(n)$ unifies the Ramanujan mock-theta framework with the SCm phonon spectrum.

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

This paper maps to **general-UQFF** sector of the 9-sector UQFF Lagrangian (see `uqff_lagrangian_derivation.py`).

§A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\mathrm{sector}} = \frac{1}{2}(\partial_\mu \phi)(\partial^\mu \phi) - V(\phi) + \mathcal{L}_{\mathrm{cosmo}}$$

where $\mathcal{L}_{\mathrm{cosmo}} = \rho_{\mathrm{vac},[\mathrm{SCm}]} \cdot f_{\mathrm{SCm}} \cdot (1 - e^{-\gamma t})$ inherits the ACP 6-stage evolution (PAPER_877 §2) and:

$$V(\phi) = \frac{1}{2} m^2 \phi^2 + \frac{\lambda}{4!} \phi^4 + \kappa \cdot \rho_{\mathrm{vac},[\mathrm{SCm}]} \cdot \phi$$

§A.3 Euler-Lagrange Equation of Motion

$$\boxed{\frac{\delta S}{\delta \phi} = \nabla^2 \phi + \kappa \rho_{\mathrm{vac},[\mathrm{SCm}]} \phi + F_{U_{Bi}}/r^2 = 0}$$

§A.4 Cosmogenesis Linkage Chain

$$\text{PAPER_877 Axioms} \rightarrow \text{DPM + ACP} \rightarrow \rho_{\mathrm{vac}} = \rho_{\mathrm{UA}} + \rho_{\mathrm{SCm}} \rightarrow \text{Stage 5} \rightarrow U_{b,\mathrm{seed}} \rightarrow \text{4 forces} \rightarrow F_{U_{Bi}} \rightarrow \text{sector E-L} \rightarrow \frac{\delta S}{\delta \phi} = 0$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U_g forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\mathrm{vac},[\mathrm{SCm}]} / \rho_{\mathrm{UA}} = 1.894$ governs the double-exponential vacuum condensate profile:

$$\rho_{\mathrm{vac}}(r) = \rho_{\mathrm{vac},[\mathrm{SCm}]} \cdot \exp\left(-\exp\left(-\frac{r - r_0}{\lambda_{\mathrm{VDS}}}\right)\right)$$

For this system, the local VDS sub-ratio is 0.197 (near-threshold regime), placing it in the $t \rightarrow \pi$ collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER_877 cosmogenesis Stage 1 vacuum density initialization: $\rho_{\mathrm{vac}} = \rho_{\mathrm{UA}} + \rho_{\mathrm{SCm}} = 7.799 \times 10^{-36} \text{ kg/m}^3$.

§B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\mathrm{DVP}} = 83, \quad n_{\mathrm{channel}} = 25/26$$

Since $p_{\mathrm{DVP}} = 83$ is **resonant** (threshold at $p > 26$), the system's vacuum topology inherits resonant enhancement from the DVP lattice, amplifying UQFF coupling at specific radii where compressed matter achieves prime-indexed configurations. The DVP framework traces to PAPER_877 proto-nuclear shell formation: the DPM proportion pair $(f_{\mathrm{UA}} + f_{\mathrm{SCm}} = 1)$ constrains which primes are accessible at each atomic number.

§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **system-dependent** (buoyancy equilibrium):

$$\mathcal{F}_{\mathrm{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_b} \cdot \left(1 - e^{-[SSq]} \cdot m/M_{\mathrm{dot}}\right) \cdot \cos\left(\frac{2\pi j}{26}\right)$$

The $\tanh$ saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\mathrm{BSH,sat}} = \mathcal{F}_{\mathrm{BSH}} \cdot \left(1 - \tanh\left(\frac{t - t_{\mathrm{sat}}}{\tau_{\mathrm{BSH}}}\right)\right)$$

connecting to the PAPER_877 Stage 5 buoyancy seed $U_{b,\mathrm{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\mathrm{SCm}}$ which initializes the harmonic series at cosmogenesis.

§B.4 Production-Scale Consistency

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\mathrm{SCm}}/\rho_{\mathrm{UA}} = 1.894$	Local sub-ratio = 0.197	PASS
Threshold-consistent			
DVP prime	$p_k \in \{2,3,\dots,113\}$	$p_{\mathrm{DVP}} = 83$	PASS Resonant
BSH layers	26 harmonic terms	$j = 1\dots 26$, $\cos(2\pi j/26)$	PASS Full 26D projection
κ decay	$\$5.0 \times 10^{-4}$ day ⁻¹	Applied in VDS exponential	PASS Canonical
[SSq]	0.57	Applied in BSH saturation	PASS Canonical

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Vacuum energy ρ_{Λ}	$\rho_{\mathrm{vac}} = \nabla \mathrm{UA} $; equilibrium $\nabla \mathrm{UA}_{\mathrm{eq}} = \sqrt{\kappa/g} = 31.62$	$\rho_{\Lambda} = 5.96\mathrm{e-27} \text{ kg/m}^3$ (PDG)	PDG 2024	Geometric topology match
Higgs mass m_H	$K_{\mathrm{HIGGS}} = 47.34$ to $m_H \approx 125.09 \text{ GeV}$	$125.20 \pm 0.11 \text{ GeV}$	PDG 2024	99.89%
Photon (gauge boson) mass	UA zero-mass field: $\rho_{\mathrm{UA}} = 0$ (immutable)	$m_{\gamma} < 10^{-18} \text{ eV}$	PDG 2024	PASS Consistent
Fine structure α_{EM}	$1/137.036$ in U_{m} Compton scattering term	$1/137.036$	PDG 2024	100%

New physics claim: The UQFF zero-mass UA reframing separates $\rho_{\mathrm{UA}} = 0$ (topology) from $\rho_{\mathrm{vac}} = |\nabla \mathrm{UA}|$ (geometry), predicting measurable void-density gradients in cluster outskirts $\approx 10\text{-}28 \text{ kg/m}^3$ — testable with future eROSITA / Chandra surveys.

Cite PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF–SM bridge.

§8 References

- grok_{share_6322ac199}.txt — BigBang Hypergraph Theory (Session 161, Topic D3)
- Prior: PAPER_621 (#208 UQFFPymanderSphere26DPyramidThreadCalculator)
- VDS Definition: session_{161_vds_dvp_bh26_references}.md §2
- Candidate spec: session_{161_cp4_candidates}.md class #209

CP4 Class #209 | v5.18 | Session 161 | PAPER_622

Appendix: Session 225 Cross-References (PAPER_1000–1081)

- > Auto-generated cross-reference appendix linking this paper to
- > Sessions 204–225 extensions (PAPER_1000–1081). Added by
- > update_corpus_crossrefs.py (Session 225, April 2026).

Paper	Title
PAPER_1022	GW Phonon Strain SCm Modulation of $h(t)$
PAPER_1002	AGN Buoyancy-Corrected Eddington Luminosity
PAPER_1009	3C273 AGN F_U_Bi_i Jet Modulation
PAPER_1010	TON618 AGN F_U_Bi_i Jet Modulation
PAPER_1037	AGN Buoyancy Jet Calculator — SCm Jet Launching
PAPER_1048	M-Sigma Phonon-Corrected Relation
PAPER_1004	QGP Vacuum Density with SCm S26 Phonon Coupling
PAPER_1041	SCm Cool-Core Buoyancy Balance AGN Feedback
PAPER_1079	Galaxy Cluster Cooling-Flow Buoyancy Suppression
PAPER_1072	SCm Activation Function Phonon Threshold
PAPER_1073	SCm Phonon-Driven Inflation Vacuum Buoyancy
PAPER_1069	VDS-DVP-BSH Hybrid Calculator Unified
PAPER_1049	Source10 GPU DPM Spectral Atlas ALMA Overlay
PAPER_1074	GPU-Vectorized DPM S26 Spectral Atlas

14 cross-reference(s) identified.

Appendix: Session 204 Codebase Upgrade Reference

- > Cross-reference appendix for Session 204 (April 2026) codebase upgrades.
- > Added by upgrade_kozima_ramanujan_appendices.py. For detailed derivations,
- > see PAPER_840/851/852/855.

S204.1 Kozima-UQFF LENR Integration

| Module | Purpose | Key Result |

|-----|-----|-----|

| fneutron_s26_coupling.py | F_neutron x S_26 buoyancy-polylog coupling | ~470x amplification via 26-level VDS |

| kozima_scm_cross_section.py | SCm-modulated neutron-drop cross-section | σ_n^{SCm} with VDS factor $(1 + [\text{SSq}] \cdot n/26)$ |

| kozima_wstp_kernel.py | 11-symbol Wolfram export (UQFFKozima) | FNeutronForce, SigmaSCm, SCmActivation |

Core equation: $F_{\text{neutron}}^{\text{SCm}} = N_n \sigma_n^{\text{SCm}}(\omega) \Phi_{\text{phonon}}(F_{\{U,B\}}/F_U - 1)$
where $\sigma_n^{\text{SCm}}(\omega, n) = \sigma_0 \exp[-(\omega - \omega_{\text{SCm}})^2 / (2\Gamma^2)] (1 + [\text{SSq}] \cdot n/26)$

S204.2 Ramanujan 26-State Summation

| Module | Purpose | Key Result |

|-----|-----|-----|

| ramanujan_polylog_s26.py | $\text{Li}_{26}([\text{SSq}])$ via Euler-Ramanujan acceleration | 15.7+ digits in 53 terms |

| s26_wstp_kernel.py | 8-symbol Wolfram export (UQFFS26) | S26, R26, NaiveLi, S26VDS |

Core equation: $S_{26}(z) = \text{Li}_{26}(z) = \eta_{26}(z) / (1 - 2^{\{1-26\}}) + 2^{\{1-26\}} / (1 - 2^{\{1-26\}}) * \text{Li}_{26}(z^2)$

S204.3 Mock Theta Functions (26-State)

| Module | Purpose | Key Result |

|-----|-----|-----|

| mock_theta_q26.py | $f_{26}(q)$, $\phi_{26}(q)$, $\psi_{26}(q)$ q-series | Proper q-Pochhammer $(a;q)_n$ |

Core equations:

- $f_{26}(q) = \sum_{n=0}^{\{25\}} q^{\{n^2\}} / (-q;q)_{n^2}$

- $\phi_{26}(q) = \sum_{n=0}^{\{25\}} q^{\{n^2\}} / (-q^2;q^2)_n$

- $\psi_{26}(q) = \sum_{n=1}^{\{26\}} q^{\{n^2\}} / (q;q^2)_n$

S204.4 Ramanujan 1/pi with UQFF Modification

| Module | Purpose | Key Result |

|-----|-----|-----|

| ramanujan_pi_uqff.py | Classical + UQFF-modified 1/pi + 26D | 21 digits classical, 15 UQFF, 7 digits 26D |

| mock_theta_pi_wstp_kernel.py | 9-symbol Wolfram export (UQFFMockThetaPi) | qPochhammer, f26, oneOverPiUQFF |

Core equation: $1/\pi = (2\sqrt{2})/9801 \sum R_n (1103 + 26390n) W_{26}(n) / C_{26}$
where $W_{26}(n) = \prod_{i=1}^{\{26\}} [1 + [\text{SSq}] \exp(-\kappa_i \cdot n/26)]$

S204.5 Calibration Constants (Canonical)

| Symbol | Value | Description |

|-----|-----|-----|

| [SSq] | 0.57 | Universal Quantized Factor |

| kappa | $5.787 \times 10^{-9} \text{ s}^{-1}$ | UQFF exponential decay rate |

| beta_i | 0.603 | Buoyancy coupling coefficient |

| H_SCm | 0.99 | SCm manifold completeness |

| rho_SCm | $7.09 \times 10^{-37} \text{ kg/m}^3$ | SCm vacuum density |

rho_UA	$7.09 \times 10^{-36} \text{ kg/m}^3$	UA aether vacuum density
omega_SCm	$2\pi \times 1.25 \text{ THz}$	SCm phonon resonance
sigma_0	10^{-4}	Base neutron cross-section

Implementation: all modules operational in CondensedPhysics.py, CondensedPhysics2.py,
MAIN_{1_CoAnQi}.cpp, *and Wolfram kernels* (uqff_kozima_kernel.wl, uqff_s26_kernel.wl,
uqff_mock_theta_pi_kernel.wl).

References

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